

Lab Exercise 4 – Task 1

Query 1

Query	house = house.
Result	true.
Explanation	Both terms are identical atoms, so they can be unified.

Query 2

Query	house = abc.
Result	false.
Explanation	house and abc are different atoms, so they cannot be unified.

Query 3

Query	edge(a,b) = edge(C,b) .
Result	C = a .
Explanation	The terms can be unified by assigning a to variable C.

Query 4

Query	edge(a,d) = edge(C,B) .
Result	C = a, B = d.
Explanation	The terms can be unified by assigning a to C and d to B.

Query 5

Query	edge(C,B) = edge(a,d) .
Result	C = a, B = d.
Explanation	The terms can be unified by assigning a to C and d to B.

Query 6

Query	$\text{edge}(A, f(g, h)) = \text{edge}(a, B)$.
Result	$A = a, B = f(g, h)$.
Explanation	The terms can be unified by assigning a to A and $f(g, h)$ to B .

Query 7

Query	$\text{edge}(A, A) = \text{edge}(b, c)$.
Result	false.
Explanation	A would need to be both b and c . Since b and c are different atoms, the terms cannot be unified.

Query 8

Query	$\text{edge}(A, A) = \text{edge}(b, b)$.
Result	$A = b$.
Explanation	Both occurrences of A match b , so the terms can be unified.

Query 9

Query	$A = B, B = f$.
Result	$A = B, B = f$.
Explanation	A is unified with B , then B is unified with f . Therefore, both variables have the value f .

Query 10

Query	$A = B, B = f, A = e$.
Result	false.
Explanation	A and B are unified, and B is assigned f , so A also becomes f . A cannot then be unified with e because f and e are different atoms.

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21 ?- house = house.  
true.  
  
22 ?- house = abc.  
false.  
  
23 ?- edge(a,b) = edge(C,b).  
C = a.  
  
24 ?- edge(a,d) = edge(C,B).  
C = a,  
B = d.  
  
25 ?- edge(C,B) = edge(a,d).  
C = a,  
B = d.  
  
26 ?- edge(A,f(g,h)) = edge(a,B).  
A = a,  
B = f(g, h).  
  
27 ?- edge(A,A) = edge(b,c).  
false.  
  
28 ?- edge(A,A) = edge(b,b).  
A = b.  
  
29 ?- A = B, B = f.  
A = B, B = f.  
  
30 ?- A = B, B = f, A = e.  
false.
```